Verified Localized Episodic Grounding for Network Analysis: A Controlled Mechanism Study

***Abstract*—Verified Localized Episodic Grounding (VLEG) combines local reflection generation, verification against labeled historical episodes, and retrieval of reflections with source evidence. Five symbolic experiments examine consolidation, grounding and refinement at 7B and 70B model scales. An additional public-flow experiment evaluates five historical-memory seeds on the same 1,000 independently partitioned, deduplicated UNSW-NB15 records using a 70B model. VLEG improves Macro-F1 over unverified first reflections by 17.02 percentage points, but underperforms raw flow retrieval: 59.68% versus 63.92%. Attack recall increases from 71.39% to 77.74%, accompanied by a false positive rate increase from 38.13% to 48.78%. The paired conditional 95% interval for the Macro-F1 difference is $[-5.91, -2.53]$ points. These results distinguish source-label checking and memory coverage from reliable downstream generalization. They support neither a universal hybrid-memory advantage nor operational intrusion-detection readiness. Historical symbolic numbers remain subject to provenance reconciliation; the completed flow experiment records exact model and source versions, prediction-level checkpoints and independently recomputed metrics.**

***Index Terms*—Episodic memory, hallucination mitigation, large language models, network traffic analysis, verified memory consolidation**

I. Introduction

A. Background: Retrieved Examples and Memory Consolidation

Large language models can use retrieved examples to support classification and analysis without parameter updates [1], [2]. In network analysis, such examples could be labeled historical flow records or incident summaries. Their usefulness depends on retaining relevant context while avoiding unsupported conclusions. This paper studies that memory-design problem using controlled symbolic tasks and a separate public-flow benchmark; it does not establish performance on operational production networks.

Persistent memory separates stored experience from the model's finite prompt window [3]–[5]. Raw-example retrieval preserves evidence, while semantic consolidation compresses episodes into summaries or heuristics [6], [7]. Compression can reduce repeated reasoning but can also remove exceptions or introduce claims that the evidence does not support.

VLEG combines local reflection generation, verification against the originating labeled episode, and retrieval of the reflection with its raw evidence. We evaluate whether these operations improve label consistency, memory coverage and downstream accuracy, and report the circumstances in which they do not. Network-flow analysis is evaluated in a separate public benchmark experiment with frozen historical-memory seeds and held-out flow fingerprints.

B. Problem Definition: Evidence and Context in Memory

Raw examples can burden inference as the number and complexity of episodes increase [8], [9]. Local reflections offer compact guidance, but a reflection can contradict its source label or cite irrelevant attributes. Feedback-based systems such as Reflexion and Voyager already use evaluation to improve later behavior [10], [11]; our focus is the narrower decision of whether to admit an individual reflection into memory after checking it against its source episode.

A global summary can represent context-dependent rules, including the background-conditioned rule used in our experiments. Heterogeneity alone does not make global consolidation potentially difficult under a restricted summary budget. The practical question is whether summarization preserves relevant conditions and exceptions within the available evidence and context budget [12], [13]. Local and global memory therefore require empirical comparison rather than an assumed ordering.

C. Verified Localized Episodic Grounding

VLEG generates a candidate reflection from an episode's attributes and checks it against the complete historical episode, including its known label. Rejected candidates receive a critique and can be revised up to a fixed budget K. Only accepted candidates are stored. The generator and verifier are separate calls to the same underlying model, not statistically independent models.

At inference, VLEG retrieves memory tuples and provides the model with both raw episodes and their accepted reflections. This hybrid context may preserve evidence while supplying a compact heuristic, but its benefit is an empirical question. The five symbolic experiments compare memory strategies, grounding modalities, iteration budgets and reflection diagnostics at 7B and 70B model scales; a sixth experiment evaluates a controlled adaptation on public flow records. They show improved label consistency and memory coverage, alongside mixed downstream results and persistent unsupported attribution. These results establish boundaries of the evaluated implementation rather than universal correctness guarantees.

D. Paper Organization

Section II reviews related memory methods. Section III defines VLEG. Section IV reports the symbolic experiments, reports the additional public-flow benchmark, and discusses limitations. Section V concludes.

II. Related Work

A. Episodic Memory and Retrieval

In-context learning uses demonstrations in the prompt to support task adaptation without parameter updates [1]. Retrieval-augmented generation connects generation to external evidence retrieved for the current input [2]. LongMem extends access to past context through a memory bank and retrieval mechanism [4]. These approaches make historical evidence available without requiring every experience to be converted into a rule. Their memory contents and training procedures differ, so they should not be treated as interchangeable implementations of a single memory architecture.

Retrieval studies address demonstration selection [14], [15] and effective context length [8], complementing broader surveys of memory design [6], [7]. These questions concern which evidence reaches inference, independently of whether that evidence is subsequently summarized.

For labeled classification tasks, raw-example retrieval offers a direct control: each historical record retains its observed features and label. It can preserve exceptions that a compressed summary omits, but its usefulness depends on selecting relevant examples within a finite context budget. A retrieved example also does not establish that a neighboring target has the same label. Retrieval relevance, evidence coverage and prediction correctness therefore require separate evaluation.

Our raw-memory baseline follows this evidence-preserving principle. It provides labeled historical episodes at inference without reflection-generation calls. The symbolic experiments compare it with abstract and hybrid context, while the public-flow experiment shares label-free feature retrieval across conditions to control example selection. This adaptation isolates the supplied memory content more closely than a comparison that changes both retrieval and context construction. It remains a controlled classification baseline, rather than a reproduction of the full training and retrieval procedures of the cited systems.

B. Local Reflection and Feedback

MemoryBank stores, retrieves and updates information from past interactions [16]. Generative agents combine an experience store with higher-level reflections [17], while Reflexion converts task feedback into verbal reflections retained for subsequent trials [10]. These works show several ways to use experience beyond storing raw examples. They differ in what constitutes an episode, which feedback is available, when memories are updated and how stored text influences later behavior.

Related work also examines procedural skill acquisition [11], structured agent memory [18], and timeline-based memory management [19]. Their different memory objectives motivate explicit controls for the classification setting studied here.

The distinction between feedback and verification is especially relevant here. Feedback can identify an unsuccessful action or prediction and motivate revision. A source-episode check instead asks whether a candidate explanation agrees with the observed record under a specified prompt. Neither mechanism, by itself, proves that the resulting explanation identifies the correct causal factors. A reflection may repair its label while retaining an unsupported threshold or treating an incidental attribute as a general condition.

VLEG studies a bounded per-episode admission procedure: generate a candidate, check it against the labeled source, revise rejected candidates, and store an accepted reflection with its evidence. Historical labels are available to the check, whereas target labels are excluded from inference. We distinguish an unverified first proposal from a feedback-based multi-attempt comparator; these provide different opportunities for correction. Both are controlled adaptations, not complete reproductions of published agent systems. The evaluation consequently separates admission coverage and source-label agreement from explanation support and downstream generalization.

C. Semantic Consolidation and Hybrid Grounding

Semantic consolidation compresses experience into hypotheses or summaries that guide subsequent reasoning. Hypothesis-Driven Reasoning iteratively generates and verifies hypotheses against episodic evidence, then combines hypotheses with retrieved episodes at inference [12]. It provides a close comparison for the unit of abstraction: hypotheses induced across episodes versus VLEG's reflections attached to individual source records. Global hypotheses can express conditional relations; indeed, one background-conditioned rule represents our synthetic environment. Heterogeneity alone therefore does not imply that global abstraction must fail.

Inductive reasoning and hypothesis generation are studied in [13], [20]–[23]; complementary work considers memory architectures [24]–[26] and iterative feedback or critique [27], [28]. We compare these research directions through abstraction scope, correction signals and inference evidence.

Hybrid grounding supplies both an abstraction and supporting examples. This retains access to concrete evidence, but can introduce conflicting guidance when the abstraction is inaccurate or applies too broadly. Its effectiveness depends on the content and applicability of the reflection, not merely the presence of both memory types. Comparing raw, abstract and hybrid context is necessary to identify these trade-offs.

VLEG's contribution is a controlled assessment of local admission checking and evidence–reflection pairing, including their limitations. The public-flow evaluation also includes raw retrieval restricted to admitted episodes, helping distinguish admission from reflection content. We report balanced classification, attack recall, false alarms and construction cost rather than inferring utility from memory coverage alone. These comparisons test when verified local memory is useful without assuming a universal advantage over direct retrieval or global consolidation.

III. The Proposed Method

A. Overview of Verified Localized Episodic Grounding (VLEG)

VLEG stores a labeled historical episode with an accepted local reflection. The implementation distinguishes generation, checking and inference roles. The following formulation describes their information access and stopping behavior; it does

not guarantee correctness of the verifier or general validity of an accepted rule.

Let $\mathcal{E} = \{e_1, \ldots, e_n\}$ be historical episodes whose labels are available during consolidation. A pure episodic memory stores those labeled episodes directly. In the measured experiments, episodes are symbolic scene tuples. A network-flow implementation instead requires actual flow features and an independently labeled historical partition. Test labels are excluded from inference in either setting.

Conversely, global semantic consolidation [12] attempts to distill $\mathcal{E}$ into a compact set of universal rules or hypotheses, $\mathcal{H}$. This can be modeled as finding a global hypothesis set that maximizes the likelihood of the observed episodes:

$$\mathcal{H}^* = \arg\max_{\mathcal{H}} P(\mathcal{H} \mid e_1, e_2, \ldots, e_n) \tag{1}$$

A single conditional hypothesis can represent the contextual rule used here. In other datasets, a limited-length summary may omit important exceptions. This is a capacity and evidence question, not a consequence that follows from heterogeneity alone.

VLEG bridges this gap by shifting the abstraction target from the global distribution to the localized episode, while enforcing strict logical verification. Rather than attempting to find a universal $\mathcal{H}^*$, VLEG constructs a memory bank consisting of verified episodic-semantic tuples:

$$\mathcal{M}_{VLEG} = \{(e_i, r_i^*) \mid e_i \in \mathcal{E},\ \mathcal{V}(r_i^* \mid e_i) = 1\} \tag{2}$$

where r_i^* is a localized semantic reflection (a detection rule or heuristic) derived specifically from episode e_i, and $\mathcal{V}(\cdot \mid e_i)$ is a strict verification function that returns 1 when the implemented verifier accepts the candidate under its checking prompt; acceptance is a model judgment and can be wrong.

The stored tuple retains the source episode alongside its accepted reflection. This permits later comparison with the evidence, although the inference model may still follow an unsupported condition. The admission gate is only as reliable as the verifier for the specified error class.

B. Framework Components: Generator, Verifier, and Memory Bank

The implementation has a reflection generator, a verifier and a memory bank. Generation and checking use separate prompts and model calls. They use the same underlying model, so shared errors and dependent responses remain possible.

1) The Reflection Generator ($\mathcal{G}_\theta$): The Reflection Generator is an LLM parameterized by θ, tasked with distilling a high-dimensional raw traffic episode e_i into a localized, natural language abstraction r_i. Formally, at iteration $j = 0$, the generator samples an initial candidate reflection conditioned solely on the raw episode:

$$r_i^{(0)} \sim \mathcal{G}_\theta(\cdot \mid e_i) \tag{3}$$

In subsequent iterations ($j > 0$), if the initial reflection is deemed invalid, the generator is conditioned on the raw episode, its previous candidate reflection $r_i^{(j-1)}$, and a natural language critique $c^{(j-1)}$ provided by the verifier:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid e_i, r_i^{(j-1)}, c^{(j-1)}) \tag{4}$$

While systems like Reflexion [10] terminate after generating $r_i^{(0)}$ and immediately commit it to memory, VLEG treats $\mathcal{G}_\theta$ strictly as a proposal distribution whose outputs are inherently untrusted until validated.

2) The Strict Verifier ($\mathcal{V}_\phi$): The verifier checks the candidate against one labeled historical episode. It can reject a mismatched label, an invented attribute or an uncheckable claim. This role is distinct from generation, but the two calls are not statistically independent.

The verifier outputs a tuple consisting of a binary validity score $v^{(j)} \in \{0, 1\}$ and a diagnostic critique $c^{(j)}$:

$$(v^{(j)}, c^{(j)}) = \mathcal{V}_\phi(e_i, r_i^{(j)}) \tag{5}$$

If $v^{(j)} = 1$, the candidate passes the implemented admission check. If $v^{(j)} = 0$, its critique is returned to the generator. Admission establishes only that this verifier accepted the reflection for this episode; it does not certify a universal rule or a lower false-positive rate than another system.

3) The Hybrid Memory Bank ($\mathcal{M}$): Once a reflection successfully passes the verifier (i.e., $v^{(j)} = 1$), it is denoted as the verified reflection r_i^*. The Hybrid Memory Bank $\mathcal{M}$ stores the paired tuple of the raw episode and its verified abstraction:

$$\mathcal{M} = \bigcup_{i=1}^{N} \left\{ (e_i, r_i^*) \mid \exists j \le K \text{ s.t. } \mathcal{V}_\phi(e_i, r_i^{(j)})_v = 1 \right\} \tag{6}$$

where K is the maximum allowed refinement iterations. If an episode fails to yield a valid reflection within K iterations, it is discarded to preserve memory integrity.

The memory bank supports retrieval of both source evidence and a local reflection. This is the representation used in the hybrid modality; raw-only and abstract-only inputs remove one of its components for comparison.

C. Learning Mechanism: Iterative Cyclic Refinement

The refinement loop revises a rejected candidate using the verifier critique. It stops at the first accepted candidate or after K attempts. Reaching the budget without acceptance discards the episode from this reflection memory.

Formally, let $e_i \in \mathcal{E}$ denote a raw, high-dimensional traffic episode. The refinement process is governed by two interacting models: the Reflection Generator G_θ and the Strict Verifier V_ϕ. At the initial iteration ($j = 0$), the generator produces a candidate reflection conditioned solely on the raw episode:

$$r_i^{(0)} \sim G_\theta(\cdot \mid e_i) \tag{7}$$

Once a candidate reflection $r_i^{(j)}$ is generated, it is passed to the verifier V_ϕ. The verifier acts as a deterministic logical gate, evaluating the reflection strictly against the grounding evidence e_i. It outputs a tuple consisting of a binary validity score $v_i^{(j)} \in \{0, 1\}$ and a natural language critique $c_i^{(j)}$:

$$(v_i^{(j)}, c_i^{(j)}) = V_\phi(e_i, r_i^{(j)}) \tag{8}$$

Note that V_ϕ's only argument besides the candidate reflection is the single episode e_i it was generated from — by construction it has no access to any other episode. This can expose a reflection that misstates e_i's own attributes or label,

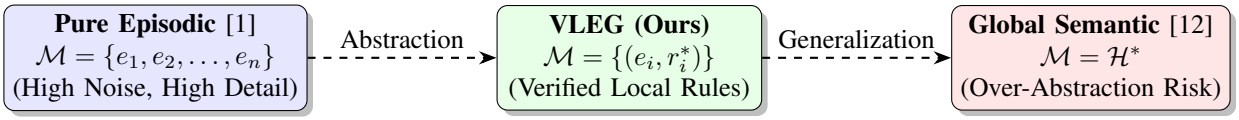


Fig. 1. The theoretical positioning of VLEG within the memory abstraction spectrum. VLEG bridges the gap between pure episodic retention of raw traffic and global semantic consolidation by pairing raw episodes with accepted local abstractions.

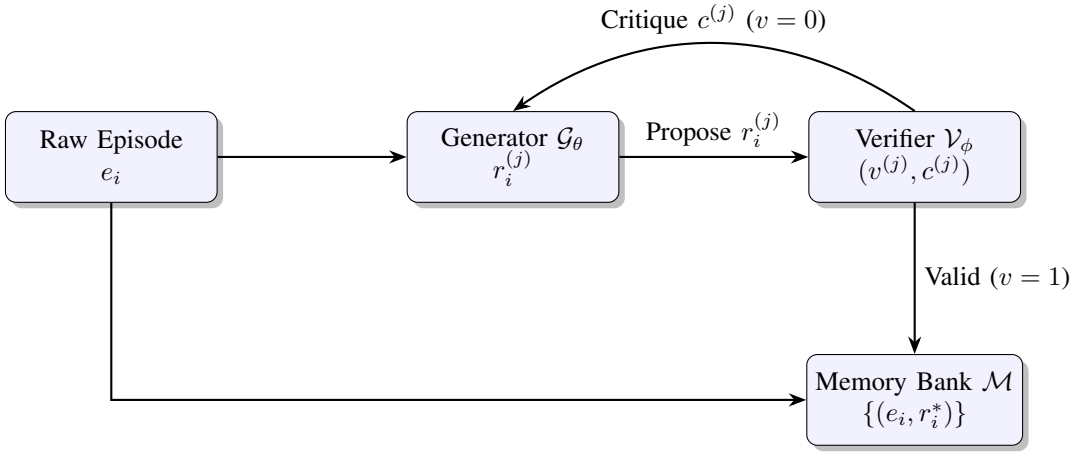


Fig. 2. Architectural flow of the VLEG framework components. Separate generator and verifier calls admit only candidates accepted by the implemented check to the Hybrid Memory Bank alongside their raw episodic counterparts.

but not sufficient to determine whether an attribute the reflection treats as necessary is actually causally relevant across episodes versus merely true of this one instance. This defines a key boundary condition of the verifier, which we investigate empirically in the memory-quality assessment (Section IV-F).

If $v_i^{(j)} = 1$, the reflection is deemed valid, the loop terminates, and the verified reflection $r_i^* = r_i^{(j)}$ is committed to the memory bank. If $v_i^{(j)} = 0$, the reflection contains a logical inconsistency or hallucination. The critique $c_i^{(j)}$ is then fed back into the generator to produce a refined reflection for the next iteration:

$$r_i^{(j+1)} \sim G_\theta(\cdot \mid e_i, r_i^{(j)}, c_i^{(j)}) \tag{9}$$

This process repeats until either $v_i^{(j)} = 1$ or a maximum iteration limit K is reached. If the limit K is exhausted without validation, the episode is deemed too ambiguous, and no reflection is stored, thereby prioritizing memory precision over recall.

Remark III.1 (Scope of the admission check). Let A_j denote reaching attempt j, E_j an erroneous candidate at that attempt, and B_j its acceptance by the verifier. For $j = 1, \ldots, K$, suppose $\Pr(E_j \mid A_j) \le p_h$ and $\Pr(B_j \mid A_j, E_j) \le p_v$ for a specified error class. Since the loop stores the first accepted candidate,

$$\begin{aligned} \Pr(\text{erroneous commitment}) &\le \sum_{j=1}^{K} \Pr(A_j \cap E_j \cap B_j) \\ &\le \min\{1, K p_h p_v\}. \end{aligned} \tag{10}$$

This union bound needs no independence assumption. It does not imply that increasing K reduces erroneous commitments, nor does it bound error conditional on admission. A geometric power of p_v would require repeated acceptance checks on the same candidate, which the implemented loop does not perform. The probabilities are not estimated by the present experiments.

A single labeled episode can expose a contradiction or a fabricated attribute, but cannot establish which attributes cause a label or whether a heuristic transfers to other episodes. the memory-quality assessment measures this distinction in the symbolic setting; public-flow explanations require additional cross-episode or human assessment.

D. Inference Integration: Hybrid Episodic-Semantic Grounding

Once the memory bank $\mathcal{M} = \{(e_i, r_i^*)\}_{i=1}^N$ has been populated with accepted tuples through the iterative cyclic refinement process, the system must effectively leverage this knowledge to solve a novel target task T (e.g., classify a held-out flow or an unseen feature combination). Traditional retrieval-augmented generation (RAG) and standard in-context learning paradigms predominantly rely on episodic-only retrieval [2], [14], injecting raw, unstructured experiences directly into the context window. While this preserves high-fidelity details, it forces the reasoning module to perform ad-hoc abstraction at test time, which frequently leads to cognitive overload and degraded performance in complex, multi-source traffic environments.

To overcome this limitation, VLEG introduces a dual-pathway integration strategy termed *Hybrid Episodic-Semantic Grounding*. This approach simultaneously injects the raw telemetry e_i and its corresponding verified semantic rule r_i^* into the context window, aligning with advanced memory integration paradigms that advocate for structured, multi-level

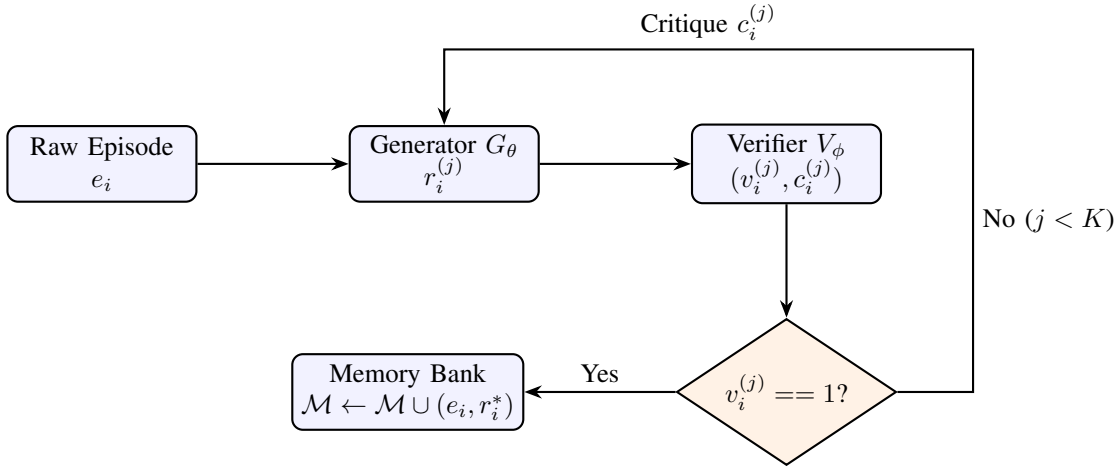


Fig. 3. State transition diagram of the Iterative Cyclic Refinement mechanism. The loop stores the first candidate accepted by the verifier to the semantic memory bank.

memory representations [6], [7]. Our empirical evaluations further characterize the *task- and scale-dependent boundary conditions* where hybrid grounding yields advantages versus when alternative grounding modalities perform comparably or better.

1) Semantic-Driven Retrieval: Given a novel target task T, the system must first retrieve the top-k most relevant memory tuples. Because the raw episodes e_i are high-dimensional and often contain task-irrelevant noise, VLEG performs retrieval primarily in the semantic space defined by the verified reflections r_i^*.

Let $\phi(\cdot)$ denote a pre-trained dense embedding function. The relevance score $S(T, \mathcal{M}_i)$ for a given memory tuple $\mathcal{M}_i = (e_i, r_i^*)$ is computed using the cosine similarity between the embedding of the target task and the embedding of the verified reflection:

$$S(T, \mathcal{M}_i) = \frac{\phi(T) \cdot \phi(r_i^*)}{\|\phi(T)\| \|\phi(r_i^*)\|} \tag{11}$$

The system then selects the subset of k tuples that maximize this similarity score:

$$\mathcal{M}_{top} = \underset{\mathcal{S} \subset \mathcal{M}, |\mathcal{S}|=k}{\arg\max} \sum_{(e_i, r_i^*) \in \mathcal{S}} S(T, (e_i, r_i^*)) \tag{12}$$

2) Hybrid Prompt Formulation: After retrieving $\mathcal{M}_{top}$, the inference module constructs a hybrid context window. The fundamental premise of hybrid grounding is that the verified reflection r_i^* provides the abstract, logical scaffold (the "what" and "why"), while the raw episode e_i provides the concrete, high-dimensional grounding (the "how"). This permits comparison with the source evidence, but does not guarantee that unsupported generalization is avoided.

The final inference prompt P is constructed by concatenating the target task T with the retrieved hybrid tuples, separated by structural delimiters:

$$\begin{aligned} Q_i &= \big[\text{Episode: } e_i \oplus \text{Verified Rule: } r_i^*\big], \\ P &= \text{PromptBuilder}\Big(\bigoplus_{(e_i, r_i^*) \in \mathcal{M}_{top}} Q_i \oplus \text{Target: } T \Big). \end{aligned} \tag{13}$$

where $\oplus$ denotes the sequence concatenation operator. The reasoning LLM, parameterized by θ, then generates the final prediction $\hat{y}$ conditioned on this hybrid prompt:

$$\hat{y} = \text{LLM}_\theta(P) \tag{14}$$

Hybrid context provides a reflection and its source evidence. Its benefit depends on whether the reflection guides the model toward relevant attributes or anchors it on a misleading condition. The grounding-modality ablation compares hybrid, raw-only and abstract-only inputs under shared retrieval.

E. Final Algorithm and Execution Flow

The Verified Localized Episodic Grounding (VLEG) framework synthesizes the previously defined components—the Reflection Generator G_θ, the Strict Verifier V_ϕ, and the Hybrid Inference Module I_ψ—into a cohesive, end-to-end execution pipeline. This pipeline systematically bridges the gap between raw episodic retention [1] and abstract semantic reasoning [12] by ensuring that only accepted local heuristics are utilized during downstream inference.

The execution flow of VLEG is formally divided into two distinct phases: **Phase 1: Memory Consolidation** and **Phase 2: Hybrid Inference**.

During Phase 1, the system ingests a stream of raw, high-dimensional traffic episodes $\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$. For each episode e_i, the Generator G_θ proposes an initial candidate reflection $r_i^{(0)}$. The Verifier V_ϕ then evaluates this candidate strictly against the grounding episode e_i, outputting a binary validity flag $v \in \{0, 1\}$ and a natural language critique c. If $v = 0$ (indicating a label-level hallucination or logical inconsistency), the critique c is fed back into G_θ to produce a refined reflection $r_i^{(1)}$. This iterative cyclic refinement continues until the reflection is validated ($v = 1$) or a predefined maximum iteration limit K is reached. If validated, the tuple (e_i, r_i^*) is committed to the Memory Bank $\mathcal{M}$. This strict gating mechanism bounds the propagation of unverified heuristics for label-contradiction errors, directly addressing the vulnerabilities of single-step reflection pipelines [10].

During Phase 2, the system is presented with a novel target task T. A Retriever R computes the cosine similarity between the task embedding and the stored memory tuples, extracting the top-k most relevant memories $\mathcal{M}_{top} \subset \mathcal{M}$. The Inference Module I_ψ is then prompted using Hybrid Episodic-Semantic Grounding, receiving both the raw episodes and their verified reflections to generate the final output. This dual-pathway integration aligns with advanced memory paradigms [6], making both representations available for the evaluated inference conditions.

The complete step-by-step procedure is formalized in Algorithm 1.

The admission gate prevents storage when the verifier rejects a candidate. Its practical reliability depends on the verifier and the error class being checked. Retaining raw evidence permits downstream comparison with a reflection, but does not ensure that the model performs that comparison correctly.

IV. Experiments

A. Experimental Settings

The symbolic evaluations use synthetic scenes with shape, color, texture, background and a deterministic size distractor. On white backgrounds, ANOMALY requires a red circle; black backgrounds invert this rule. Texture and size are non-causal. Level 1 contains eight white-background combinations; Level 2 contains sixteen across both backgrounds. Both evaluate combinations used to form memory. Level 3 withholds four from memory and tests all sixteen; masked-only accuracy measures withheld-combination generalization. Seeds change the withheld subset, not the underlying population.

Reported symbolic runs use Qwen2.5-7B-Instruct or Llama-3.3-70B-Instruct, nomic-embed-text embeddings and temperature 0. Generator and verifier are separate calls to the same model. Generation excludes historical labels; verification receives labeled source episodes, so refinement uses supervised feedback. Target labels are reserved for scoring. We vary consolidation, grounding modality and $K \in \{1, 2, 3, 5\}$, measuring accuracy, memory coverage, label consistency, attribution flags and construction tokens. The grounding-modality ablation shares retrieval across modalities. First-proposal reflections, feedback-based Reflexion and global hypotheses are controlled adaptations; their implementation differences are identified below.

The 7B and latest 70B archives have been reconciled, but lack original verifier replies and exact model digests. They precede strict boolean-parser repairs and were not rerun. Cross-scale comparisons also change model family and cannot isolate parameter count. Repeated queries across seeds are not independent samples, and temperature 0 does not guarantee identical outputs. the public-flow benchmark separately specifies the public-flow data, frozen model versions, split, metrics and runtime costs.

B. Controlled Memory-Strategy Comparison

We compare VLEG, global-hypothesis consolidation, unverified single-step reflection, and raw in-context retrieval on the three symbolic task levels. The comparisons characterize

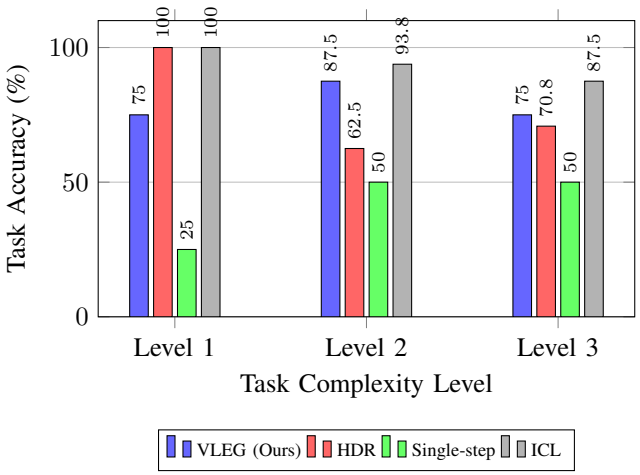


Fig. 4. Measured Task Accuracy (%) across three levels of task complexity (reduced-order reimplementation, Qwen2.5-7B-Instruct; Level 3 pooled across 3 masking seeds, $n = 48$). ICL leads at every level; HDR is bimodal (perfect at Level 1's non-conditional rule, weak once the rule becomes background-conditional); Single-step tracks each level's ANOMALY base rate rather than discriminating between scenes.

this small environment and these adaptations; they do not establish superiority over published systems on their original benchmarks.

Within each model-scale comparison the same underlying model and temperature are used. VLEG uses $K = 3$. Level 1 has eight queries and Level 2 sixteen. Level 3 is pooled over three masking seeds, giving 48 evaluations of sixteen recurring combinations. Repeat runs matched reported accuracies in the checked conditions, which is an empirical repeatability observation rather than a determinism guarantee.

The reported 7B accuracies are ICL 100.0%, 93.8% and 87.5% at Levels 1–3, versus VLEG 75.0%, 87.5% and 75.0%. Global-hypothesis consolidation recovers the uniform-background rule at Level 1 (100.0%), but has pooled accuracy 70.8% at Level 3. The historical single-step reflection comparator scores 25.0%, 50.0% and 50.0%; the inspected reflections all predict ANOMALY, making its accuracy equal to the corresponding anomaly base rate. These values describe the historical adaptation, not the changed multi-attempt feedback baseline now present in the repository.

At 7B, raw retrieval performs best in this environment. This limits the evidence for an advantage from consolidation at that scale. The grounding-modality ablation isolates grounding content more directly, and the limitations below explain why this globally expressible oracle does not establish a general weakness of global consolidation.

The reported 70B check evaluates Levels 1 and 2 once each and Level 3 across three masking seeds. The latest archived, feedback-baseline revision gives Level-3 pooled accuracies of VLEG 91.67% (44/48), raw retrieval 89.58% (43/48), global hypothesis 83.33% (40/48), and feedback-based Reflexion 81.25% (39/48). VLEG exceeds raw retrieval in two seeds but is lower in the third. These replace the earlier 93.75% VLEG and single-step-baseline numbers; the feedback-based comparator is not the same condition as the 7B first-proposal comparator. This reverses the descriptive ordering at 7B.

Algorithm 1 Verified Localized Episodic Grounding (VLEG)

Input: Episodic stream $\mathcal{E} = \{e_1, \ldots, e_n\}$, Target Task T, Max Iterations K, Retriever R

Initialize Memory Bank $\mathcal{M} \leftarrow \emptyset$
// Phase 1: Memory Consolidation
for each $e_i \in \mathcal{E}$ **do**
  $j \leftarrow 0$
  $c \leftarrow \emptyset$ ▷ Initialize empty critique
  $v \leftarrow 0$ ▷ Initialize validity flag to Invalid
  while $v = 0$ **and** $j < K$ **do**
    $r_i^{(j)} \leftarrow G_\theta(e_i, c)$ ▷ Generate or refine reflection
    $(v, c) \leftarrow V_\phi(e_i, r_i^{(j)})$ ▷ Verify reflection against raw episode
    $j \leftarrow j + 1$
  end while
  if $v = 1$ **then**
    $\mathcal{M} \leftarrow \mathcal{M} \cup \{(e_i, r_i^{(j-1)})\}$ ▷ Store verified tuple
  end if
end for
// Phase 2: Hybrid Inference
$\mathcal{M}_{top} \leftarrow R(\mathcal{M}, T, k)$ ▷ Retrieve top-k relevant tuples
Context $\leftarrow T \cup \mathcal{M}_{top}$ ▷ Construct hybrid prompt
Output $\leftarrow I_\psi$(Context) ▷ Generate final reasoning answer
return Output

Because the model family also changes and the evaluation population is small, the difference should not be attributed to parameter count alone.

C. Component Isolation (Grounding Modality Ablation)

This ablation isolates which retrieved content is shown to the inference model: the raw episode, the accepted reflection, or both. The same retrieved tuples are used across conditions to reduce retrieval-quality confounding.

To achieve this, we evaluate the framework under three distinct ablation conditions. The first condition is *Hybrid Grounding (VLEG)*, which represents the complete proposed methodology where the reasoning module is conditioned on both the retrieved raw episodes e_i and their corresponding verified reflections r_i^*. The second condition is *Abstract-Only Grounding*, where the reasoning module is provided solely with the verified reflections r_i^*, completely omitting the raw traffic episodic data. The third condition is *Raw-Only Grounding*, which serves as a baseline mirroring standard In-Context Learning (ICL) [1], where the model receives only the raw episodes e_i without any distilled semantic guidance.

To isolate the effect of grounding *content* from retrieval quality, all three conditions retrieve the same top-k memory tuples (via cosine similarity on the verified reflection's embedding, as VLEG's memory bank is built the same way regardless of which fields are shown at inference); only which fields of each retrieved tuple are shown to the reasoning LLM differs between conditions.

The measured results, shown in Figure 5, do not match the hypothesized ordering. On Level 2 (no masking), Hybrid Grounding is indeed the best condition (87.5% vs. 81.2% for both alternatives). But on Level 3 — the condition this experiment was specifically designed to stress-test — the ordering inverts: Hybrid Grounding is the *worst* of the three conditions, both overall (73.8% pooled across 5 masking seeds, $n = 80$, vs. 78.8% for Abstract-Only and 75.0% for Raw-Only) and specifically on the masked combinations that never appeared during memory formation (60.0% pooled, $n = 20$, vs. 70.0% for both alternatives). Hybrid never outperformed both alternatives on Level 3 in any of the 5 seeds tested. We connect this to the memory-quality assessment (Section IV-F): verified reflections frequently cite causally irrelevant attributes as if they were determinative (see the empirical caveat in Section III), and pairing a raw episode with such a reflection appears to anchor the reasoning model on the reflection's stated — sometimes spurious — condition rather than letting it reason more freely from the raw examples, which is more costly precisely when generalization to an unseen combination is required.

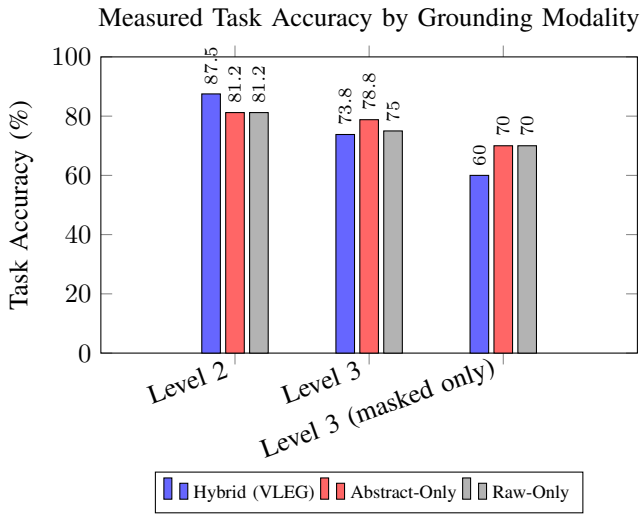


Fig. 5. Measured Task Accuracy across grounding modalities (reduced-order reimplementation, Qwen2.5-7B-Instruct). Level 3 figures are pooled across 5 masking seeds ($n = 80$ overall, $n = 20$ for the masked-only subset). Hybrid Grounding wins on Level 2 but is the worst condition on Level 3, especially on the masked combinations it was intended to help with — the opposite of the original hypothesis.

At 7B, abstract-only context exceeds hybrid context by 5.0 percentage points overall and 10.0 points on the masked subset. The result does not support a claim that hybrid context is necessary for this task. The reported 70B check below has a different overall ordering.

We repeated this experiment with Llama-3.3-70B-Instruct across the same 5 masking seeds. Unlike the memory-strategy, refinement and memory-quality comparisons, where the 7B result either reversed cleanly (memory-strategy and refinement comparisons) or stayed exactly as limited (the memory-quality assessment), this one produced a genuinely mixed picture. Pooled over 5 seeds ($n = 80$ overall, $n = 20$ masked-only): overall accuracy, Hybrid (95.0%) now ties Raw-Only (95.0%) for best, with Abstract-Only worst (91.25%) — the opposite of Abstract-Only's 7B standing. But on the masked-combo subset specifically, Abstract-Only remains the best modality (85.0% vs. 80.0% for both alternatives) — the same direction as the 7B finding, just with a narrower margin (5 points versus 10). Per-seed masked accuracy for Abstract-Only was noisy ($100\%, 75\%, 50\%, 100\%, 100\%$ across the 5 seeds), with one seed accounting for most of its overall shortfall. We do not read this as either a clean confirmation or a clean reversal of the 7B finding: the specific claim that Hybrid is the worst modality overall does not hold at 70B, but the narrower claim that Abstract-Only retrieval has some edge specifically on unseen, masked combinations persists in direction across both scales, just less dramatically.

D. Iterative Refinement Ablation

The refinement ablation tests whether increasing the refinement budget recovers rejected episodes and changes prediction accuracy. It does not assume that iteration or verification is necessary in every task.

Both $K = 1$ and $K = 3$ run the same generate-and-verify procedure. At $K = 1$, the one rejected proposal leads to discarding the episode; at $K = 3$, the generator can revise after critique. This is a refinement-budget ablation, not a verified-versus-unverified comparison. Both use the same model and hybrid inference.

Across five masking seeds, $K = 3$ achieves 75.0% overall accuracy ($60/80$) versus 63.75% ($51/80$) for $K = 1$, a difference of 11.25 percentage points. Masked-only accuracy is 60.0% ($12/20$) versus 50.0% ($10/20$). The masked-only difference is zero in three seeds and positive in two. These counts are repeated evaluations under different memory masks, not eighty independent scenes.

The mechanism behind the gain is memory bank *coverage*, not reasoning quality per reflection: across the 5 seeds, $K = 1$'s memory bank retained only 42–58% of training episodes (an episode whose first-guess reflection is rejected is discarded outright, since j is never $< K{=}1$ again), while $K = 3$'s retained 83–100%, because the critique from a rejected first attempt typically reveals the correct label and lets the generator correct itself on the second or third try. Verification at $K = 3$ is thus better understood here as *recovering episodes that a single unlucky guess would otherwise lose*, not as improving the causal quality of the reflections that do get committed — see the memory-quality assessment (Section IV-F) for a direct measurement showing no such quality improvement.

At 70B, the reported mean overall difference is 26.2 percentage points (standard deviation 8.1) across the same five seeds, and $K = 3$ admits all twelve historical episodes in each seed. The results are consistent with coverage recovery through label-informed critique; they do not show that the accepted explanations identify causal attributes more reliably.

E. Parameter Sensitivity and Overhead Analysis

The primary objective of the parameter-sensitivity study is to systematically analyze the convergence behavior of the VLEG verification loop and to quantify the associated computational trade-offs. While the iterative generate-and-verify mechanism is designed to strictly filter out hallucinated heuristics, each refinement cycle inherently incurs additional computational costs during the memory consolidation phase. Therefore, it is critical to evaluate how the strictness of the verification process impacts both the downstream reasoning performance and the overall token efficiency of the system.

To conduct this sensitivity analysis, we manipulate the maximum refinement iterations parameter, K, across a discrete set of values: $K \in \{1, 2, 3, 5\}$. The condition $K = 1$ effectively disables the iterative refinement loop, reducing the system to a single-step reflection generator akin to standard localized reflection baselines [10]. For each configuration of K, we evaluate the model on both Level 2 and Level 3 of the Experience-Transfer task. We record the resulting Task Accuracy (%) alongside the Computational Overhead, which is measured as the total number of prompt and completion tokens consumed per consolidated episodic memory.

The measured results, shown in Figure 6, contradict both claims originally hypothesized for this experiment. First, Task Accuracy does not plateau around $K = 3$. On Level 2 — a single run, verified deterministic by an exact repeat — accuracy keeps climbing substantially from $K{=}3$ (87.5%) to $K{=}5$ (93.8%), a further $+6.3$ point gain with no sign of leveling off in this range. On Level 3, we tested 3 independent masking seeds: only one of them (seed $= 42$) shows the exact plateau signature (identical bank composition, token count, and accuracy at $K{=}3$ and $K{=}5$); the other two show $K{=}5$ either matching or beating $K{=}3$ by a further $+6$–12.5 points. We do not have a reliable stopping criterion at $K = 3$ in this environment — it happens to be sufficient for some random draws of the masked subset and insufficient for others.

Second, Computational Overhead does not scale linearly with K. It is concave: the token cost per consolidated episode rises sharply from $K{=}1$ to $K{=}2$ ($+279$ tokens/episode on Level 2) and then far more slowly for each additional unit of K budget ($+71$ tokens/episode from $K{=}2$ to $K{=}3$, $+35$ from $K{=}3$ to $K{=}5$ despite that step covering two extra iterations, not one). This pattern was consistent across all seeds tested on both levels. The mechanism is straightforward: most episodes converge within 1–2 iterations regardless of the ceiling K; raising the ceiling only adds cost for the small minority of episodes that are still unresolved at iteration 3, so the marginal cost of a higher K budget shrinks as K grows even as the

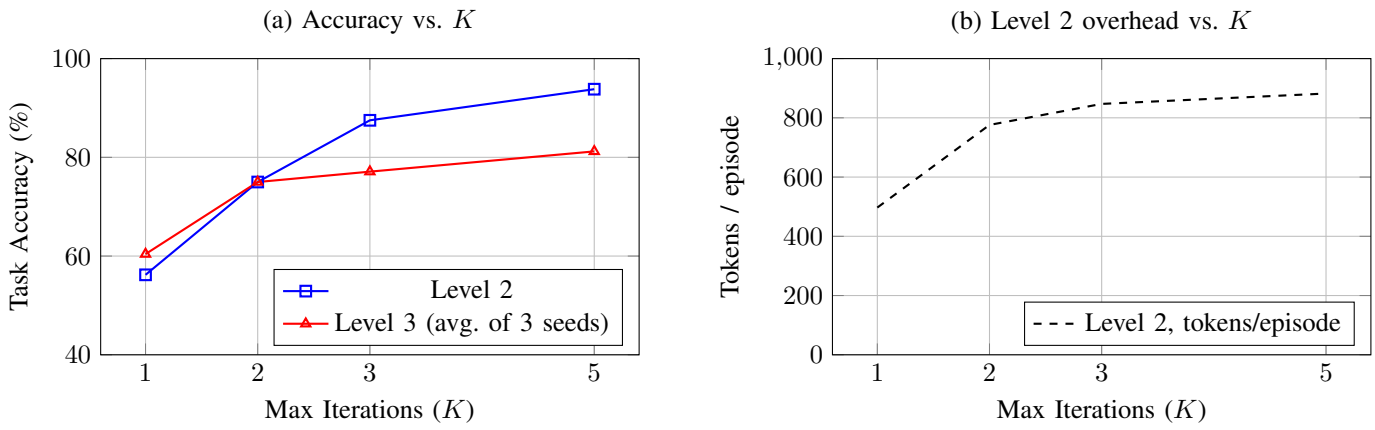


Fig. 6. Measured parameter sensitivity of the maximum refinement iterations K (reduced-order reimplementation, Qwen2.5-7B-Instruct). (a) Task Accuracy does *not* plateau at $K = 3$: Level 2 (a single deterministic run, confirmed identical on repeat) keeps improving through $K = 5$; Level 3 (averaged over 3 masking seeds) plateaus in only 1 of 3 seeds individually. (b) Token overhead per consolidated episode is concave, not linear: most of the cost increase happens between K=1 and K=2, with sharply diminishing marginal cost for extending the budget further, since only a minority of episodes ever use the additional iterations.

marginal *benefit* (per the accuracy panel) does not reliably shrink in the same way. Practitioners should not treat $K = 3$ as a validated efficiency-optimal cutoff on the basis of this experiment; whether a higher K is worth its (modest, sub-linear) additional cost depends on how much the specific downstream task benefits from the extra recovered episodes, which the refinement ablation (Section IV-D) suggests is real but seed-dependent in magnitude.

At 70B, the reported Level-3 sweep uses three masking seeds. In each, $K = 2$, $K = 3$ and $K = 5$ yield the same admitted memory, token consumption and accuracy. No episode needs a third attempt in these runs. The observed plateau is therefore at $K = 2$ in this model and task, while the 7B sweep is less uniform. This does not establish a universal stopping budget.

Concurrent GPU workloads make the affected timing measurements unsuitable for a clean overhead comparison. One reported repeat at Level 2, $K = 3$, changes the token count from 12,541 to 12,625 while preserving accuracy. Accordingly, temperature 0 is recorded as a decoding setting, and repeatability and hardware contention must be assessed empirically for each run.

F. Diagnostic Assessment of Memory Quality

The memory-quality assessment separates source-label consistency from unsupported attribution. These diagnostics complement downstream accuracy and test what a per-episode admission check actually filters in this symbolic domain.

We report admitted-memory coverage, source-label consistency and a keyword-based unsupported-attribution diagnostic. Coverage is the proportion of episodes admitted within $K = 3$; it is not meaningful as a verification metric for an always-commit comparator. Label consistency is scored against the known synthetic label. The attribution flag detects references to texture or size, which are irrelevant to the oracle. It can flag a mere mention as well as an asserted condition, so it is a coarse diagnostic. Earlier joint LLM judgments are excluded from the reported table because they conflated these axes.

Table I reports 100% source-label consistency among admitted VLEG reflections at both levels, versus 41.7–50.0% for the unverified single-step comparator. Conditional on label correctness, unsupported-attribution rates are 86.7% versus 87.5% at Level 2 and 83.3% versus 80.0% at Level 3. These small-sample rates show no clear verification-specific reduction. Source-label consistency is an observed diagnostic outcome, not a tautological guarantee of an imperfect model-based verifier.

In this diagnostic, the admitted reflections match their source labels, but frequently cite attributes that do not affect the synthetic oracle. For example, asserting that striped texture is necessary cannot be validated from a single striped instance. Cross-episode contrast is needed to distinguish an observed property from a general condition. A keyword flag is a coarse attribution diagnostic, not a causal explanation assessment for natural traffic.

At 70B, reconstructing the source order prescribed by the experiment script gives admitted-reflection label consistency of 100.0%, versus 50.0% for first proposals. This pairing is not independently attested by a checkpoint. Unconditional texture/size keyword rates are equal for both strategies: 12.5% (2/16) at Level 2 and 16.7% (2/12) at Level 3, compared with approximately 80–87% at 7B. The reduction occurs for both strategies, so it is not evidence of a verification-specific benefit. Given the small counts, equality should not be described as proof of statistical equivalence. Cross-scale grounding results are reported separately in Section IV-C.

G. Public-Flow Benchmark Validation

This experiment tests whether local verification and hybrid memory improve classification of independent public flow records. It uses UNSW-NB15, a captured testbed benchmark containing normal and generated attack traffic [29]. It is not a production deployment or zero-day detection study. The completed archive contains 500 pilot predictions and 25,000 formal predictions; an offline audit reproduces the metrics,

TABLE I
DIAGNOSTIC ASSESSMENT OF MEMORY QUALITY (EXPERIMENT 5, REDUCED-ORDER REIMPLEMENTATION, QWEN2.5-7B-INSTRUCT). LABEL CORRECTNESS AND SPURIOUS-ATTRIBUTION RATE ARE COMPUTED DETERMINISTICALLY AGAINST GROUND TRUTH, NOT VIA LLM JUDGMENT. "SPURIOUS (OF CORRECT)" = % OF LABEL-CORRECT REFLECTIONS CONTAINING A TEXTURE/SIZE KEYWORD; THE FLAG DOES NOT DISTINGUISH MENTION FROM CAUSAL ASSERTION.

Level	Strategy	Validity	Label-correct	Spurious (of correct)
Level 2	Single-Step	N/A	50.0% (8/16)	87.5% (7/8)
	VLEG (K=3)	**93.8%**	**100.0% (15/15)**	86.7% (13/15)
Level 3	Single-Step	N/A	41.7% (5/12)	80.0% (4/5)
	VLEG (K=3)	**100.0%**	**100.0% (12/12)**	83.3% (10/12)

conditional bootstrap intervals and construction-cost totals from the stored records.

The source partitions contain 175,341 historical and 82,332 test records. The frozen feature representation uses nineteen fields: duration, protocol, service, state, directional packet and byte counts, rate, TTL, load, inter-packet timing, jitter and mean packet size. IPs, timestamps, row identifiers, attack names and target labels are excluded from model inputs and retrieval keys. The public mirror is pinned to a commit; its file names are reversed relative to the official partition sizes, so role mapping is recorded explicitly and checked by row count, size, MD5 and SHA256. Bytes agree with an independent research deposit, although the original author-hosted files were not directly accessible.

Identical selected-feature vectors sometimes have conflicting labels. We exclude every row in such groups rather than choosing a label: 6,896 historical rows in 273 groups and 2,912 test rows in 105 groups. We also remove 79,746 and 33,161 remaining same-label duplicate rows, respectively, leaving 88,699 historical and 46,259 test fingerprints. A further 1,044 test fingerprints overlap the cleaned historical source and are excluded. Thus the evaluated population is deduplicated and unambiguous in this representation; removing these cases can change difficulty and class balance, so these results do not estimate performance on the original flow-frequency distribution. Session-level independence cannot be established because this partition lacks session-group metadata.

Each of five memory seeds (20261020–20261024) selects sixteen benign and sixteen malicious historical examples. They share the same 1,000 test fingerprints: 663 benign and 337 malicious. The 100 pilot queries do not overlap these formal queries. We use Llama-3.3-70B (70.6B parameters, Q4_K_M), nomic-embed-text (137M, F16), temperature 0, $K = 3$, $k = 3$, and a 32,768-token chat-context ceiling. The run records exact model digests, source/code hashes and Python 3.13.13; the server has two A800 80GB GPUs. Generator and verifier are separate calls to the same model. Historical labels are available to verification and the raw baseline, but target labels are used only after inference for scoring. The generator's first proposal receives only historical flow features.

The five conditions are raw flow retrieval, an unverified first reflection, VLEG, raw flow retrieval restricted to VLEG's admitted examples, and a single global summary of the labeled historical memory. All local conditions share a label-free raw-feature embedding retrieval pass and the same selected top-three candidates. A rejected candidate is dropped without replacement. The raw-on-admitted control therefore removes reflection content while retaining the same admission mask as VLEG. This shared raw retrieval deliberately differs from the reflection-driven formulation in Section III and tests consolidation/admission under controlled retrieval; it does not evaluate every component of the full formulation. Global-summary inference sees a corpus summary rather than the same top-three examples, and is a memory-architecture comparison rather than an equal-evidence modality ablation. These are controlled adaptations, not complete published Reflexion/HDR reproductions.

Table II and Figure 7 show that raw retrieval has the highest mean Macro-F1 among the evaluated conditions (63.92%), while VLEG reaches 59.68%. VLEG increases attack recall from 71.39% to 77.74%, but increases the false positive rate from 38.13% to 48.78%. Its accuracy falls from 65.08% to 59.74%. A post hoc always-benign reference has accuracy 66.30%, Macro-F1 39.87% and zero attack recall; this illustrates why accuracy alone is insufficient here. None of these conditions establishes deployment-quality intrusion detection, and no comparison against a trained conventional intrusion classifier is made.

Verification is beneficial relative to the unverified first-reflection condition: Macro-F1 improves by 17.02 percentage points, and attack recall rises from 6.17% to 77.74%. However, high admitted-memory coverage (98.75%, 158/160 historical episodes) does not establish reliable cross-flow explanations. Raw-on-admitted retrieval nearly matches the raw baseline (63.70% versus 63.92% Macro-F1), whereas adding the accepted reflections reduces its mean score to 59.68%. These comparisons are consistent with reflection content changing the decision trade-off; they do not establish a specific anchoring or causal-attribution mechanism. A subsequent deterministic explanation screen finds source-label agreement in 158/158 admitted VLEG reflections versus 84/160 first proposals. Explicit conjunctive conditions can be evaluated in 157/158 VLEG reflections; all 157 hold on their source flow, but 113 have at least one contrary-label match among the other archived historical draws. The remaining expression uses ambiguous $\gg$ syntax and is unresolved. The screen uses 160 unique historical flows and no test queries. It checks literal predicates, not natural-language causal validity, and the historical pool is neither independent nor representative. A human review with field-semantic judgments and adjudication remains incomplete.

A paired bootstrap resamples the same 1,000 fingerprints

TABLE II
PUBLIC-FLOW PERFORMANCE, MEAN ± SAMPLE STANDARD DEVIATION ACROSS FIVE HISTORICAL-MEMORY SEEDS (PERCENT). ALL SEEDS USE THE SAME 1,000 TEST FINGERPRINTS. INVALID OUTPUTS COUNT AS ERRORS.

Condition	Accuracy	Macro-F1	Attack recall	False positive rate
Raw flow retrieval	65.08 ± 2.93	63.92 ± 2.54	71.39 ± 8.44	38.13 ± 6.85
Unverified single-step	61.38 ± 1.08	42.66 ± 1.35	6.17 ± 1.43	9.89 ± 2.03
VLEG	59.74 ± 2.03	59.68 ± 2.03	77.74 ± 5.69	48.78 ± 1.67
Raw on admitted flows	64.86 ± 2.96	63.70 ± 2.68	71.16 ± 8.71	38.34 ± 6.61
Global summary	50.22 ± 12.45	47.60 ± 14.13	87.12 ± 22.27	68.54 ± 24.20

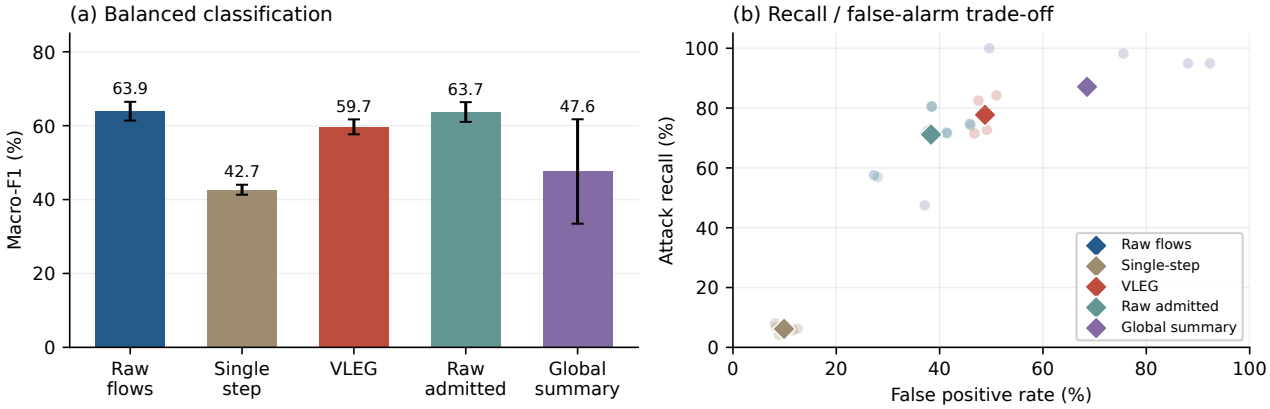


Fig. 7. Public-flow results across five memory seeds. Left: mean Macro-F1 with sample-standard-deviation bars. Right: attack recall versus false positive rate; faint points are individual seeds and large markers are means. The larger recall of VLEG is accompanied by more false alarms.

synchronously across methods and all five memory seeds (1,000 replicates). The mean VLEG-minus-raw Macro-F1 difference is -4.24 percentage points, with a conditional 95% interval of $[-5.91, -2.53]$; VLEG is lower in four seeds and slightly higher in one. VLEG-minus-single-step is $+17.02$ points, interval $[13.71, 20.39]$. VLEG-minus-raw-on-admitted is -4.02 points, interval $[-5.79, -2.28]$. These are unadjusted descriptive intervals conditional on the fixed memory draws and an assumption of independent fingerprints; they do not quantify uncertainty over new networks, sessions, model weights or memory-seed populations. Five repeated evaluations do not create 5,000 independent test flows. Per-attack recalls are archived, but rare categories include only one Backdoor, one Worms and three Shellcode cases; their percentages cannot support precise category-specific conclusions.

Table III separates one-time memory construction from query chat time. VLEG uses a mean 40,094 construction tokens and 419.43 seconds, versus 9,405 tokens and 126.54 seconds for first-proposal reflections. Raw storage needs no construction chat calls; embedding cost is separate and embedding tokens are not included. VLEG query input averages 909 tokens versus 780 for raw retrieval. Methods run in the fixed order raw, single-step, VLEG, raw-on-admitted, global summary. Consequently, prompt caching and order confound latency comparisons: the shorter observed VLEG chat time is not evidence of a causal speed advantage. Moreover, among 4,968 raw/raw-on-admitted pairs with identical reconstructed prompts, 24 predictions differ despite temperature 0. This variation limits interpretation of the small raw-versus-admission-control gap. There are 29 invalid single-step outputs and 30 invalid VLEG outputs across 5,000 predictions each; all other conditions have zero invalid outputs. False positive rate counts explicit malicious predictions on benign flows, so invalid-output counts and benign recall must be considered alongside it.

The admission-control condition shares VLEG's construction process and its cost; it does not obtain the mask for free. The completed experiment therefore supports a bounded result: checking first reflections improves on an unverified adaptation, but accepted hybrid reflections do not improve balanced classification over raw retrieval on this cleaned public-flow population.

H. Limitations

The symbolic evaluations use small populations, while the public-flow benchmark evaluates a cleaned public-flow benchmark. The latter is a controlled adaptation with shared raw retrieval, not a production deployment or a complete validation of reflection-driven retrieval. Levels 1 and 2 evaluate seen combinations; Level 3 combines seen and withheld combinations. Their accuracy must not be interpreted as held-out network generalization. The flow experiment is restricted by deduplication/conflict exclusion, missing session metadata, one model and test partition, fixed method order and incomplete human explanation audits.

The synthetic rule is expressible as a single background-conditioned statement. It tests conditional rule learning, not an impossibility of global abstraction. The 7B and 70B

TABLE III
OBSERVED PUBLIC-FLOW COSTS, MEANS ACROSS FIVE MEMORY SEEDS. CONSTRUCTION TOKENS INCLUDE INPUT AND OUTPUT. CHAT LATENCY EXCLUDES SHARED RETRIEVAL AND IS CONFOUNDED BY FIXED METHOD ORDER/CACHE REUSE.

Condition	Construction calls	Construction tokens	Construction (s)	Query mean (s)	Query p95 (s)
Raw flow retrieval	0	0	0	2.12	2.67
Unverified single-step	32.0	9,405	126.54	2.23	2.75
VLEG	120.4	40,094	419.43	1.74	2.68
Raw on admitted flows	120.4	40,094	419.43	0.95	1.19
Global summary	1.0	5,947	23.21	1.21	1.30

comparisons also change model family and cannot isolate scale causally. Observed rank reversals and equal diagnostic rates are descriptive findings in these samples, not general superiority or equivalence claims.

Verification sees a known historical label and tests a candidate against one source episode. It can reject inconsistencies detected by the model, but agreement does not establish causal attribution or cross-episode validity. Reflection diagnostics based on known synthetic attributes cannot be transferred unchanged to natural traffic, where ground-truth explanations are generally unavailable.

Finally, the 7B figures are linked to the archived initial reimplementation snapshot, while updated 70B comparisons use the latest archived feedback-baseline revision. Stored item-level accuracies were independently recounted, but these historical files omit exact model digests and original verifier replies. Boolean type validation was also repaired in the oracle-audit parser; without original replies its historical impact cannot be quantified. The symbolic evaluations were not rerun, while the public-flow benchmark is independently audited against its archived predictions and frozen source. Context limits, quantization, model digests, CPU/GPU offloading and concurrent workloads require run-specific recording. The completed public-flow run records source/model/code hashes and hardware, but does not resolve every limitation of the historical experiments.

V. CONCLUSIONS

VLEG checks local reflections before memory admission and combines accepted reflections with source episodes during inference. In the reported symbolic experiments, refinement increases memory coverage and admitted reflections are more consistent with historical labels. Downstream benefits depend on the model and task: raw retrieval leads at 7B, while the latest archived 70B comparison gives a one-correct-evaluation pooled advantage to VLEG over raw retrieval. Hybrid grounding retains a disadvantage on some withheld combinations, and per-episode verification does not show a separate benefit for unsupported attribution.

The public-flow experiment strengthens this bounded interpretation. At 70B, VLEG improves Macro-F1 over unverified first reflections but is lower than raw retrieval, while increasing both attack recall and false alarms. High memory admission coverage therefore does not establish reliable cross-flow reasoning. Further work should test calibration and cross-episode validation on new, independently reserved data, complete explanation audits, and control timing/repeatability before making deployment claims. Historical symbolic files have now been inventoried against immutable 7B and 70B Git snapshots, and 48 stored item-level accuracy records reproduce their summaries. The latest feedback-baseline results replace stale 70B comparisons. Git provenance cannot recover missing model digests or raw verifier replies, so corrected-parser reruns and human review remain required.

REFERENCES

[1] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. J. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei, "Language Models are Few-Shot Learners," *ArXiv*, vol. abs/2005.14165, 2020.

[2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., 2020, pp. 9459–9474.

[3] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as Operating Systems," *ArXiv*, vol. abs/2310.08560, 2023.

[4] W. Wang, L. Dong, H. Cheng, X. Liu, X. Yan, J. Gao, and F. Wei, "Augmenting Language Models with Long-Term Memory," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[5] Y. Wang, Y. Gao, X. Chen, H. Jiang, S. Li, J. Yang, Q. Yin, Z. Li, X. Li, B. Yin, J. Shang, and J. McAuley, "MEMORYLLM: Towards Self-Updatable Large Language Models," in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235, 2024, pp. 50 453–50 466.

[6] Y. Wu, S. Liang, C. Zhang, Y. Wang, Y. Zhang, H. Guo, R. Tang, and Y. Liu, "From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs," *ArXiv*, vol. abs/2504.15965, 2025.

[7] Z. Zhang, X. Bo, C. Ma, R. Li, X. Chen, Q. Dai, J. Zhu, Z. Dong, and J.-R. Wen, "A Survey on the Memory Mechanism of Large Language Model based Agents," *ArXiv*, vol. abs/2404.13501, 2024.

[8] C. An, J. Zhang, M. Zhong, L. Li, S. Gong, Y. Luo, J. Xu, and L. Kong, "Why Does the Effective Context Length of LLMs Fall Short?" *ArXiv*, vol. abs/2410.18745, 2024.

[9] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the Middle: How Language Models Use Long Contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[10] N. Shinn, F. Cassano, A. Gopinath, K. R. Narasimhan, and S. Yao, "Reflexion: language agents with verbal reinforcement learning," in *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

[11] G. Wang, Y. Xie, Y. Jiang, A. Mandlekar, C. Xiao, Y. Zhu, L. Fan, and A. Anandkumar, "Voyager: An Open-Ended Embodied Agent with Large Language Models," *Transactions on Machine Learning Research*, 2024.

[12] A. K. Agarwal and M. Yamada, "Hypothesis-Driven Reasoning for Large Language Models," in *Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (AAAI-26)*, 2026, pp. 1686–1693.

[13] K. He, M. Zhang, S. Yan, P. Wu, and Z. Chen, "IDEA: Enhancing the Rule Learning Ability of Large Language Model Agent through Induction, Deduction, and Abduction," in *Annual Meeting of the Association for Computational Linguistics*, 2024.

[14] Y. Zhang, S. Feng, and C. Tan, "Active Example Selection for In-Context Learning," *ArXiv*, vol. abs/2211.04486, 2022.

[15] K. Purohit, V. V, S. Bhattacharya, and A. Anand, "Sample Efficient Demonstration Selection for In-Context Learning," in *Forty-second International Conference on Machine Learning*, 2025.

[16] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang, "MemoryBank: Enhancing Large Language Models with Long-Term Memory," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 17, pp. 19 724–19 731, 2024.

[17] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative Agents: Interactive Simulacra of Human Behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.

[18] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang, "A-MEM: Agentic Memory for LLM Agents," *ArXiv*, vol. abs/2502.12110, 2025.

[19] S. H. Kim, K. T. Ong, T. Kwon, N. Kim, K. Ka, S. Bae, Y. Jo, S. Hwang, D. Lee, and J. Yeo, "Towards Lifelong Dialogue Agents via Timeline-based Memory Management," in *North American Chapter of the Association for Computational Linguistics*, 2024.

[20] S. Kim, P. Phunyaphibarn, D. Ahn, and S. Kim, "Playgrounds for Abstraction and Reasoning," in *NeurIPS 2022 Workshop on Neuro Causal and Symbolic AI (nCSI)*, 2022.

[21] K. il Lee, H. Koh, D. Lee, S. Yoon, M. Kim, and K. Jung, "Generating Diverse Hypotheses for Inductive Reasoning," *ArXiv*, vol. abs/2412.13422, 2024.

[22] L. Qiu, L. Jiang, X. Lu, M. Sclar, V. Pyatkin, C. Bhagavatula, B. Wang, Y. Kim, Y. Choi, N. Dziri, and X. Ren, "Phenomenal Yet Puzzling: Testing Inductive Reasoning Capabilities of Language Models with Hypothesis Refinement," in *International Conference on Representation Learning*, B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, Eds., vol. 2024, 2024, pp. 1452–1481.

[23] Y. Zhou, H. Liu, T. Srivastava, H. Mei, and C. Tan, "Hypothesis Generation with Large Language Models," in *Proceedings of EMNLP Workshop of NLP for Science*, 2024.

[24] T. Kim, M. Cochez, V. François-Lavet, M. Neerincx, and P. Vossen, "A machine with short-term, episodic, and semantic memory systems," in *Proceedings of the Thirty-Seventh AAAI Conference on Artificial Intelligence and Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence and Thirteenth Symposium on Educational Advances in Artificial Intelligence, AAAI'23/IAAI'23/EAAI'23*, 2023.

[25] J. Li and J. Li, "Memory, Consciousness and Large Language Model," *ArXiv*, vol. abs/2401.02509, 2024.

[26] Y. Wang and X. Chen, "MIRIX: Multi-Agent Memory System for LLM-Based Agents," *ArXiv*, vol. abs/2507.07957, 2025.

[27] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegreffe, U. Alon, N. Dziri, S. Prabhumoye, Y. Yang, S. Gupta, B. P. Majumder, K. Hermann, S. Welleck, A. Yazdanbakhsh, and P. Clark, "Self-Refine: Iterative Refinement with Self-Feedback," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[28] Z. Gou, Z. Shao, Y. Gong, Y. Shen, Y. Yang, N. Duan, and W. Chen, "CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing," in *International Conference on Learning Representations*, 2024.

[29] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Military Communications and Information Systems Conference (MilCIS)*, 2015.